IIMT3642: Introduction to Managing and Mining Big Data

Zhepeng (Lionel) Li, PhD
Associate Professor
Innovation and Information Management

Role of Data in Modern Business

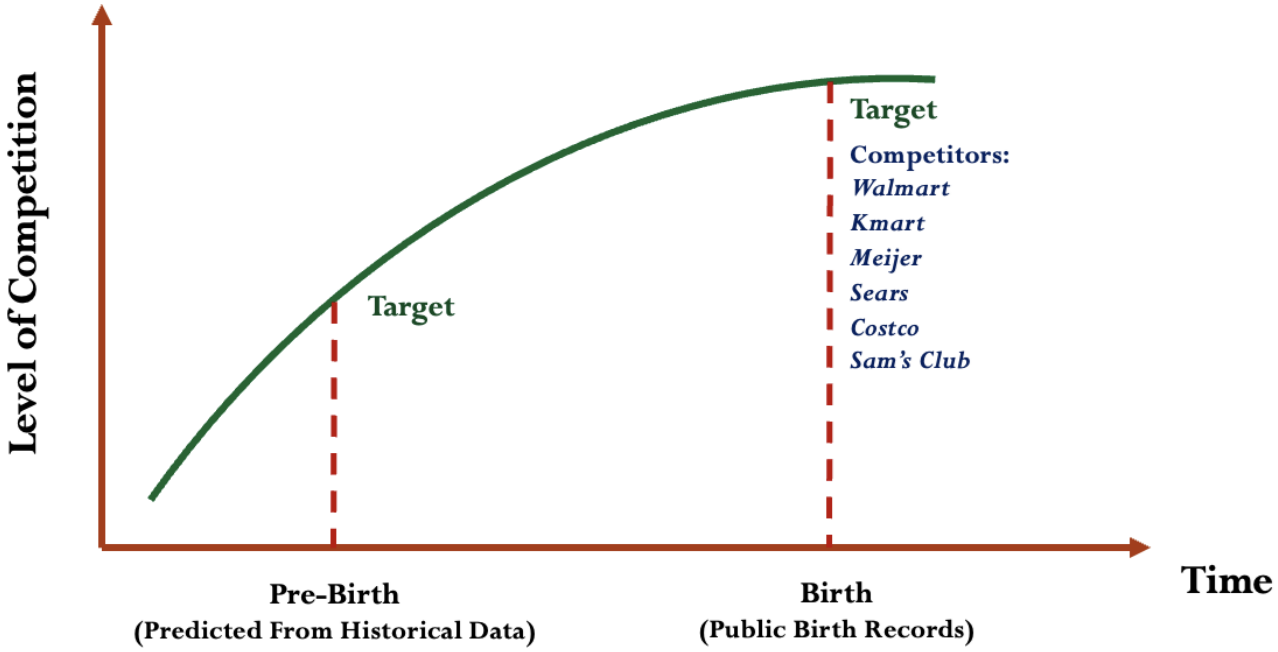
- Transactions and Routine Operations are supported by data
  - Retailing Business
    - data about product price, quantity, credit card number, and transaction time etc.
    - For example: grocery shopping/cross sales
  - Online Banking
    - data about customer, transactions, account number, account balance etc.
    - For example: risk management/personal credit loans
  - Healthcare Management
    - data about patients (e.g. age, gender, weight etc.) , test results (e.g. blood test), and medical history etc.
    - For example: diagnosis/readmission(mortality) forecasting

- Data as By-products
  - Large amount of data are generated and stored during business and human activities
  - However, data are once considered by-products of computer-based information systems
  - Data are primarily viewed as historical records
    - Support of system operating/recovery
    - Source of summary statistics, etc.
- Historical records ≠ Actionable Insights

- Data as Strategic Assets
  - Modern business calls to connect historical records with future actions
  - Turning data from by-products to strategic assets through data-driven decision making (DDDM)
  - Data-Driven Decision Making: the practice of basing decisions on the analysis of data, rather than purely on intuition.
  - Data Mining Techniques are valuable in DDDM

- One example is Target in retail business
  - Target is a US-based retail store that sells products to families with young kids.
  - They know that “As soon as we get them (customers) buying diapers from us, they’re going to start buying everything else too”.
  - Common practice: obtain birth information from PUBLIC birth records to send out special offers to new parents.

- Time matters



A study of 32 major corporations across a variety of industries (Davenport 2006) shows that corporations with leading capabilities of data analytics are clear leaders in their respective industries.

Davenport, T. H. 2006. “Competing on Analytics,” Harvard Business Review (January 2006) 99-107.

- Other business dimensions

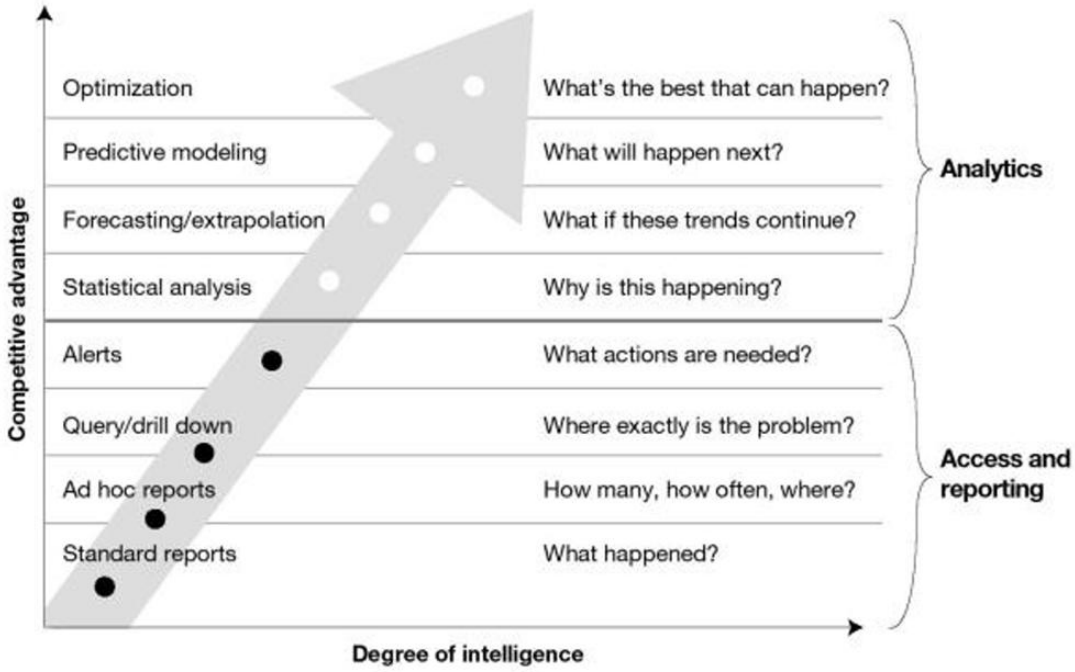
FUNCTION	DESCRIPTION	EXEMPLARS
Supply chain	Simulate and optimize supply chain flows; reduce inventory and stock-outs.	Dell, Wal-Mart, Amazon
Customer selection, loyalty, and service	Identify customers with the greatest profit potential; increase likelihood that they will want the product or service offering; retain their loyalty.	Harrah's, Capital One, Barclays
Pricing	Identify the price that will maximize yield, or profit.	Progressive, Marriott
Human capital	Select the best employees for particular tasks or jobs, at particular compensation levels.	New England Patriots, Oakland A's, Boston Red Sox
Product and service quality	Detect quality problems early and minimize them.	Honda, Intel
Financial performance	Better understand the drivers of financial performance and the effects of nonfinancial factors.	MCI, Verizon
Research and development	Improve quality, efficacy, and, where applicable, safety of products and services.	Novartis, Amazon, Yahoo

- Tangible Results
  - “Data-driven decisions are better decisions – it’s as simple as that. Using big data enables managers to decide on the basis of evidence rather than intuition.” (Harvard Business Review, Oct. 2012)
  - “Data should be at the heart of strategic decision making in businesses. Data leads to insights; business owners and managers can turn those insights into decisions and actions that improve the business.” (Forbes, June 2016)
  - “Companies using data-driven decision making are 5% more productive and 6% more profitable those who do not” (MIT study on 330 North American Companies in 2012)
  - “Successful application of big data analytics can generate $300 billion annually for US health care” (McKinsey Report 2011)

Relationship with Database and Summary Statistics

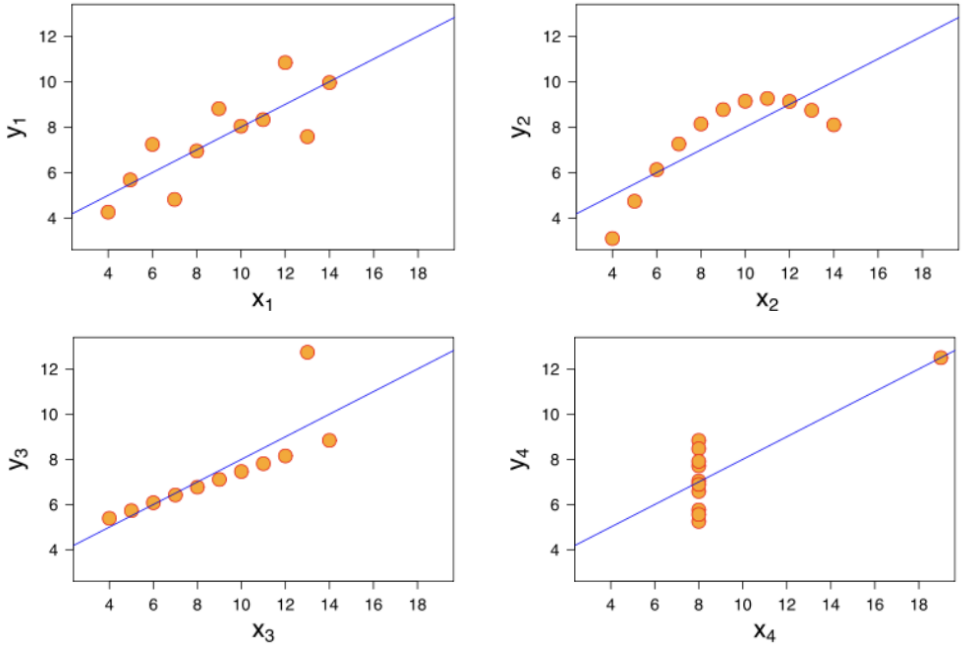
- Looking Beyond Database Queries

Decision makings based on simple queries of databases do NOT create as much competitive advantages and values as decision makings based on deep analysis of data



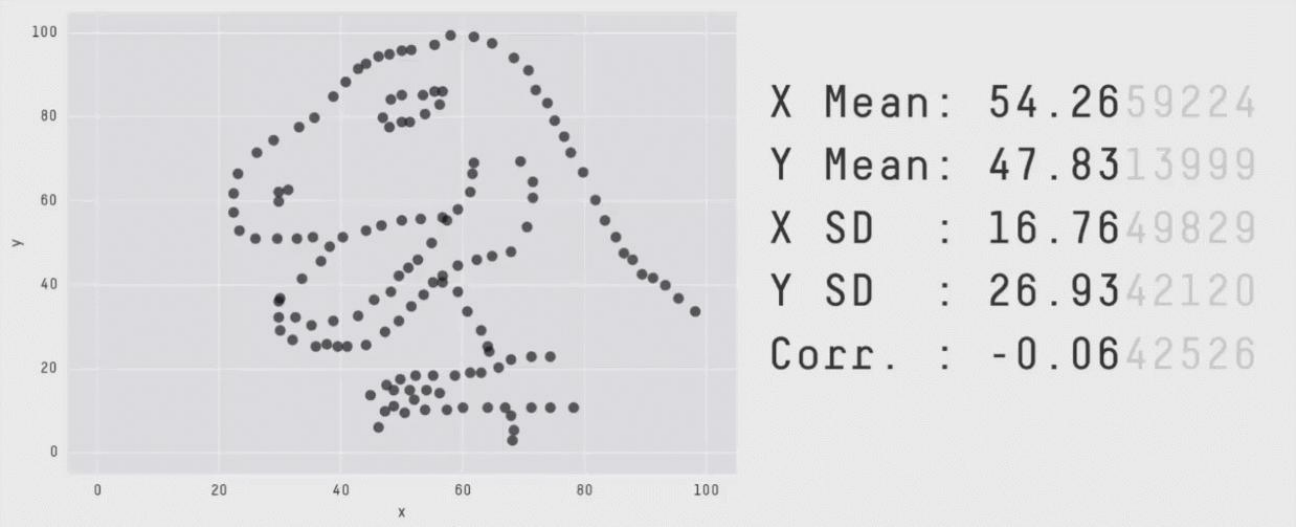
Davenport, T. H., Harris, J. G. 2007. "Competing on analytics: The new science of winning," Harvard Business School Press.

- Looking Beyond Summary Statistics
  - One Classic example: Anscombe's Quartet - Four scatterplots of data points that have nearly identical descriptive statistics (mean, variance, correlation) look drastically different



- Looking Beyond Summary Statistics
  - Simply relying on statistics could be misleading

It is possible to generate bivariate data with a given mean, median, and correlation in any shape you like - even a dinosaur!



https://www.research.autodesk.com/publications/same-stats-different-graphs/
https://www.scientificamerican.com/article/what-this-graph-of-a-dinosaur-can-teach-us-about-doing-better-science/

Data Terminology

Data

Information

Data Mining

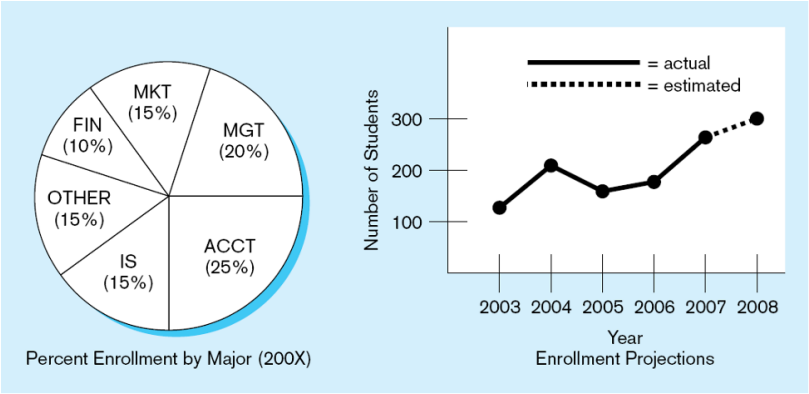
Statistics

BIG Data

Machine Learning

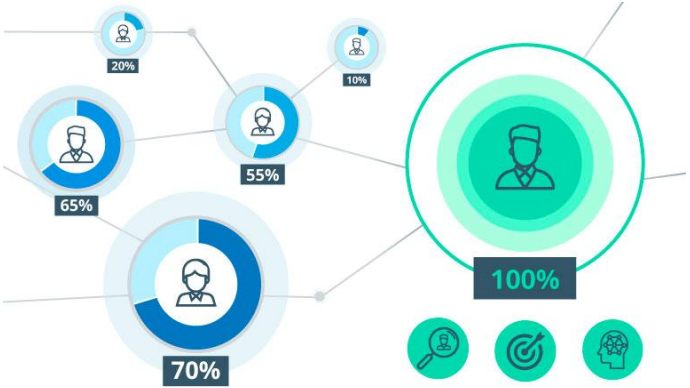
- Data
  - Data: raw facts concerning objects and events
  - Examples of Data
    - Mark's GPA is 2.9.
    - The current enrollment of Accounting major is 300.
    - The number of customers who churned in last month is 393.
  - Data by themselves do not convey much meaning

- Information
  - Information: data processed to carry a meaning
  - Examples of Information



  - Information should be meaningful by putting facts into a context

- Knowledge
  - Knowledge: a collection of insights that can guide actions
  - Examples of Knowledge
    - In the previous example, more seats are needed due to the projected number of enrollment
    - For a different example, say the likelihood of each customer to churn when certain conditions are met



  - Knowledge is actionable

DATA
Database Queries
Information

DATA
Data Mining
Knowledge

- Data Mining vs. Machine Learning
  - Data Mining: is the automatic process that discovers knowledge from data
    - Apply developed method/model/algorithm to extract actionable knowledge
  - Machine Learning: is the study of programmable approach that improve automatically through experience (Mitchell 1997).
    - Develop new method/model/algorithm for different learning tasks

- Big Data
  - Big Data: is characterized by volume, velocity, variety
    - Volume: total amount of digital data created worldwide would grow to 180 zettabytes by 2025 (IDC 2014)
    - Velocity: speed of data in and out
    - Variety: structured and unstructured data (text, audio, video etc.)
  - Implications of Big Data
    - Storage
    - Data Management
    - Data Computing

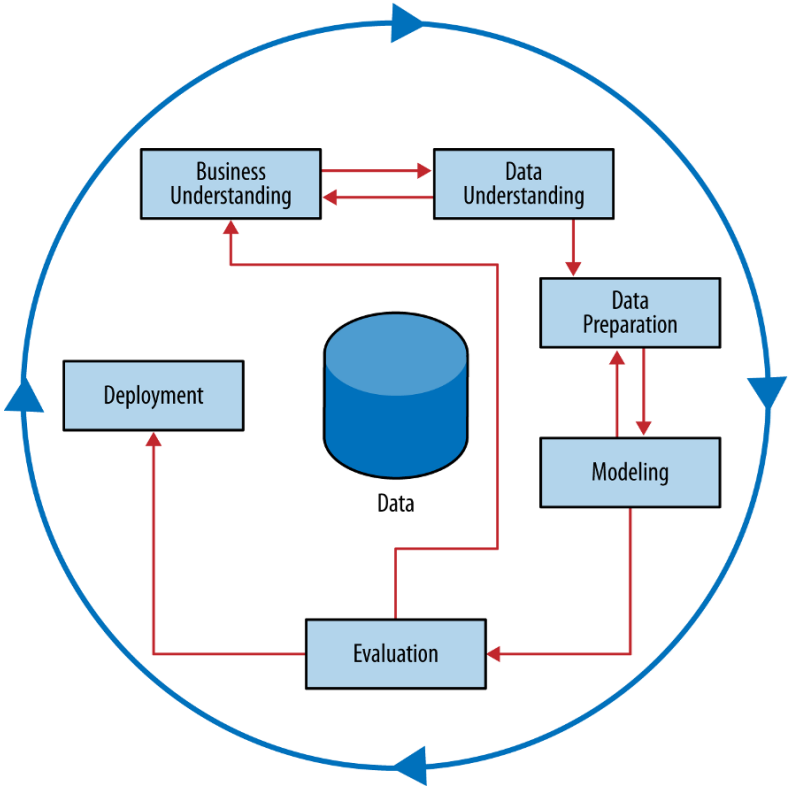
- Statistics
  - Statistics have one emphasis on explanatory (e.g., hypothesis testing) while Data Mining primarily focuses on predictive purposes
  - For example,
    - Is the churn rate (12%) this month significantly lower than that (14%) in last month? If so, why; if not, why? ← Statistics
    - What will be the churn rate in the next month? ← Data Mining

Data Mining Process CRISP

- Business Understanding: accurately define the business problem to be solved; clearly define the success metrics (e.g., accuracy, value, cost etc.)

- Data Understanding: what data need to be collected (internal, external)?

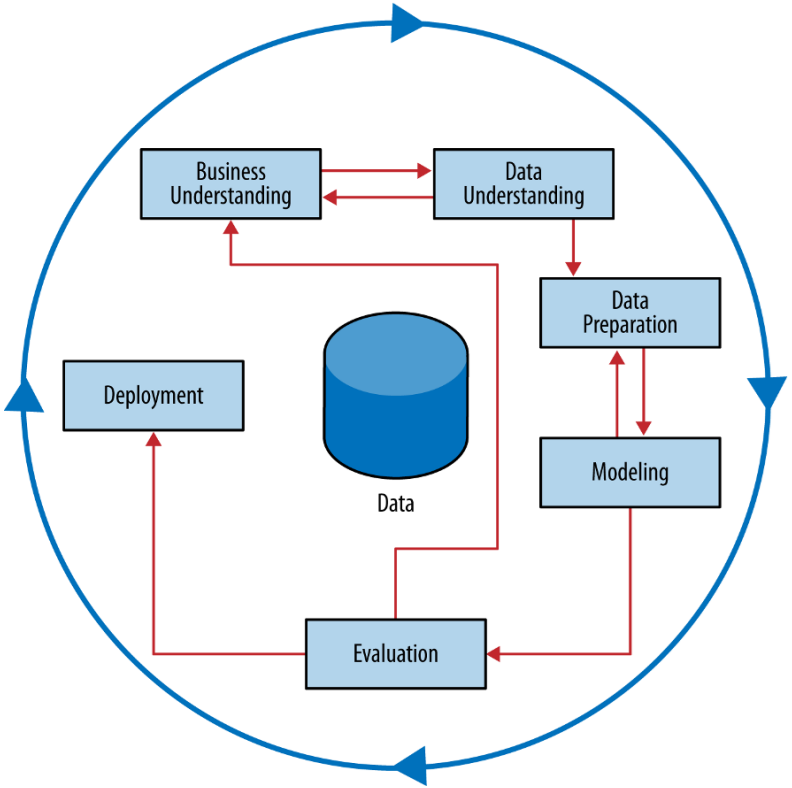
- Data Preparation: data transformation (e.g., scaling, discretization), missing value etc.



- Modeling: designing suitable data mining methods to solve the business problem.

- Evaluation: how to evaluate?

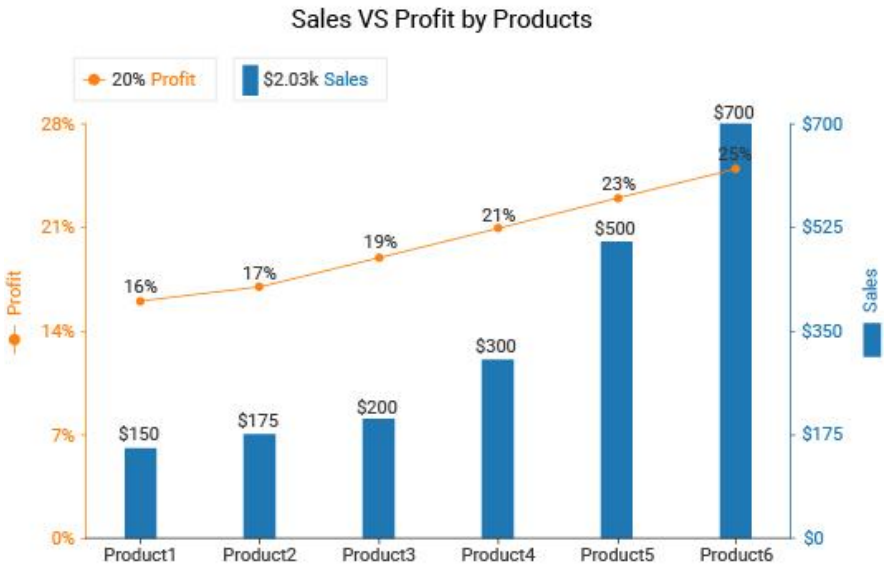
- Deployment: put your designed data mining system into real use to create value for an organization.



Data-analytic Thinking (Descriptive vs. Predictive)

- Descriptive Analytics: Understanding the Past
  - **Definition:**
    - Descriptive analytics summarizes historical data to provide meaningful insights.
  - **Purpose:**
    - To understand past performance and identify trends and patterns.
  - **Techniques:**
    - Summary statistics (mean, median, mode, standard deviation)
    - Data visualization (histograms, bar charts, line graphs)
    - Data aggregation and reporting

- **Example:**
  - Analyzing sales data to determine the best-performing products over the last quarter.



- Techniques in Descriptive Analytics
  - Summary Statistics:
    - Measures of central tendency (mean, median, mode)
    - Measures of dispersion (range, variance, standard deviation)
  - Data Visualization:
    - Histograms: Show frequency distribution of a single variable.
    - Bar Charts: Compare different categories.
    - Line Graphs: Track changes over time.
    - Pie Charts: Display proportions of a whole.
  - Data Aggregation:
    - Combining data from multiple sources to provide a summary.

- Predictive Analytics: Forecasting the Future
  - **Definition:**
    - Predictive analytics uses historical data and statistical algorithms to predict future outcomes.
  - **Purpose:**
    - To identify trends, forecast future events, and inform decision-making.
  - **Techniques:**
    - Regression analysis
    - Time series analysis
    - Machine learning algorithms (e.g., decision trees, neural networks)
  - **Example:**
    - Predicting customer churn rates based on historical behavior data.

- Techniques in Predictive Analytics
  - Regression Analysis:
    - Linear Regression: Models the relationship between variables.
    - Logistic Regression: Models the probability of a binary outcome.
  - Machine Learning Algorithms:
    - Decision Trees: Splits data into branches to make predictions.
    - Random Forests: Ensemble method combining multiple decision trees.
    - Neural Networks: Models complex relationships in data.

Aspect	Descriptive Analytics	Predictive Analytics
Definition	Analyzes historical data to understand what has happened.	Uses historical data and statistical algorithms to forecast future events.
Focus	What happened?	What could happen?
Techniques	Data aggregation, data mining, statistical analysis.	Regression analysis, time series analysis, machine learning.
Tools Used	Excel, Tableau, Power BI, SQL.	R, Python (scikit-learn, TensorFlow), SAS, IBM SPSS.
Applications	Business reporting, customer segmentation, market analysis.	Risk management, customer retention, sales forecasting.
Outcome	Provides insights into past performance and trends.	Predicts future trends and behaviors based on past data.

Data-analytic Thinking (Supervised vs. Unsupervised)

- Supervised Learning
  - **Definition:**
    - Supervised learning is a type of machine learning where the model is trained on a labeled dataset, meaning that each training example is paired with an output label.
  - **Key Techniques:**
    - Classification: Predicting a categorical label (e.g., spam detection, loan approval).
    - Regression: Predicting a continuous value (e.g., house prices, stock prices).
  - **Process:**
    - Training: The model learns from a labeled dataset.
    - Validation: The model is tested on a separate validation set to tune parameters.
    - Testing: The final model is evaluated on a test set to assess its performance.
  - **Applications:**
    - Customer churn prediction.
    - Fraud detection.
    - Medical diagnosis.

- Unsupervised Learning
  - Definition:
    - Unsupervised learning is a type of machine learning where the model is trained using data that does not have labeled responses.
  - Key Techniques:
    - Clustering: Grouping data points into clusters based on similarity (e.g., customer segmentation).
    - Association Rules: Finding interesting relationships between variables in large databases (e.g., market basket analysis).
    - Dimensionality Reduction: Reducing the number of random variables under consideration (e.g., PCA).
  - Process:
    - Data Collection: Gather unlabeled data.
    - Exploration: Analyze the data to understand its structure.
    - Modeling: Apply algorithms to find hidden patterns and relationships.
  - Applications:
    - Market segmentation.
    - Anomaly detection.
    - Recommender systems.

Ethical Considerations

Ethical considerations are crucial in data mining tasks to ensure that data is used responsibly and that individuals' rights are respected. Here are some key ethical considerations to keep in mind:

- Privacy and Confidentiality
  - **Data Collection:** Ensure that data is collected with the informed consent of individuals. Avoid collecting more data than necessary.
  - **Data Anonymization:** Anonymize personal data to protect the identities of individuals. Be aware that re-identification can sometimes be possible with enough data.
  - **Data Access:** Limit access to sensitive data to authorized personnel only. Use encryption and other security measures to protect data.
- Informed Consent
  - **Transparency:** Clearly inform individuals about what data is being collected, how it will be used, and who it will be shared with.
  - **Opt-In/Opt-Out Options:** Provide individuals with the option to opt-in or opt-out of data collection and use.

- Data Accuracy and Integrity
  - **Data Quality:** Ensure that the data used is accurate, complete, and up-to-date. Poor data quality can lead to incorrect conclusions and decisions.
  - **Bias and Fairness:** Be aware of and mitigate potential biases in the data and algorithms. Ensure that the data mining process does not reinforce existing biases or create new ones.
- Transparency and Accountability
  - **Explainability:** Make the data mining process and the resulting models as transparent as possible. Ensure that decisions made by algorithms can be explained and justified.
  - **Accountability:** Establish clear lines of responsibility for data mining tasks. Ensure that there are mechanisms in place to address any issues or harm caused by data mining activities.
- Ethical Use of Data
  - **Purpose Limitation:** Use data only for the purposes for which it was collected. Avoid using data in ways that could harm individuals or groups.
  - **Non-Discrimination:** Ensure that data mining tasks do not lead to discrimination against individuals or groups based on race, gender, age, or other protected characteristics.

Introduction to Google Colaboratory

- What is Google Colaboratory?

  Google Colaboratory, often referred to as Google Colab, is a cloud-based platform that allows you to write and execute Python code in a Jupyter Notebook environment. It is a free service provided by Google Research and is particularly popular in the fields of data science, machine learning, and artificial intelligence.

- Key Features
  - **Cloud-Based:** Since Colab is cloud-based, you don't need to install anything on your local machine. All computations are performed on Google's cloud servers.
  - **Integration with Google Drive:** You can easily save and share your notebooks on Google Drive. This makes it convenient to collaborate with others.
  - **Pre-Installed Libraries:** Colab comes with many popular Python libraries pre-installed, including TensorFlow, Keras, PyTorch, and Pandas, among others.
  - **Easy Sharing and Collaboration:** Notebooks can be shared just like Google Docs. Multiple users can collaborate in real-time.
  - **Rich Text Editing:** You can combine code with rich text elements (e.g., headings, lists, links, images) to create well-documented and visually appealing notebooks.
  - **Free Access to GPUs and TPUs:** Google Colab provides free access to powerful hardware accelerators like GPUs (Graphics Processing Units) and TPUs (Tensor Processing Units), which can significantly speed up machine learning tasks.